Week 07 — Spectral Density & Spectral Representation

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The big picture. The spectral density $\mathcal{S}(f)$ is the Fourier transform of the autocovariance: it distributes the process variance across frequencies (Herglotz, spectral representation).

So far we described a stationary process in the *time domain* through its ACVS $\{\gamma_\tau\}$. This week we cross into the *frequency domain*: the very same second-order information is re-read as a distribution of the process variance across frequencies. This is the "second face" of a stationary process. From now on we fix the sampling interval $\Delta = 1$ and work with ordinary frequency f (not angular frequency $\omega = 2\pi f$, which only scatters factors of 2π).

Key definitions

Definition — Spectral density function (SDF)

For a stationary process $\{X_t\}_{t\in\mathbb{Z}}$ whose ACVS is **summable** ($\{\gamma_\tau\} \in \ell^1$), the SDF is the (discrete-time) Fourier transform of the ACVS:

$$\mathcal{S}(f) = \sum_{\tau\in\mathbb{Z}} \gamma_\tau \, e^{-2\pi i f \tau}, \qquad f \in \mathbb{R}.$$

It is inverted back to the ACVS by $\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f)\, e^{2\pi i f \tau}\, df$; in particular

$$\gamma_0 = \mathrm{Var}(X_t) = \int_{-1/2}^{1/2} \mathcal{S}(f)\, df.$$

A valid SDF is **real**, **even** ($\mathcal{S}(-f) = \mathcal{S}(f)$), **non-negative** ($\mathcal{S} \geq 0$) and **1-periodic**. (In continuous time, replace the sum by $\mathcal{S}(f) = \int_{\mathbb{R}} \gamma(\tau) e^{-2\pi i f \tau}\, d\tau$.)

What $\mathcal{S}(f)$ actually says

$\mathcal{S}(f)\, df$ = amount of variance ("power") carried by the components with frequency near f, averaged over realisations. The SDF slices the total variance γ_0 frequency by frequency. A *flat* spectrum = white noise (every frequency equally present, hence "white"); a *peaked* spectrum = a strong oscillation at that frequency.

Definition — Harmonic (sinusoidal) process

With $A > 0$ and $\Theta \sim \mathrm{Unif}(-\pi, \pi)$ independent, set $X_t = A\cos(\nu t + \Theta)$ with oscillation frequency ν. Then $\mathbb{E}[X_t] = 0$ and $\mathrm{Cov}(X_{t+\tau}, X_t) = \frac{1}{2}\mathbb{E}[A^2]\cos(\nu\tau)$. It is stationary, but its ACVS **does not decay**: $\{\gamma_\tau\} \notin \ell^1$, so it has **no ordinary SDF**. Its spectrum is a *pair of lines* (jumps) at $\pm\nu/(2\pi)$.

Why we need more than the SDF

The SDF (Def. 1) assumes $\{\gamma_\tau\} \in \ell^1$. The harmonic process shows that a perfectly legitimate stationary process can violate this. We therefore need a more general object that *always* exists: the integrated spectrum.

Definition — Integrated spectrum $S^{(I)}$

A function $S^{(I)}$ on $[-\frac{1}{2}, \frac{1}{2}]$ that is right-continuous, bounded, **non-decreasing**, with $S^{(I)}(-\frac{1}{2}) = 0$ and $S^{(I)}(\frac{1}{2}) = \sigma^2 = \gamma_0$. It behaves like a (scaled) cumulative distribution function in frequency and represents the ACVS as

$$\gamma_\tau = \int_{-1/2}^{1/2} e^{2\pi i\tau f}\, dS^{(I)}(f).$$

When $\{\gamma_\tau\} \in \ell^1$, $S^{(I)}$ has a density, which is exactly the SDF: $\mathcal{S} = dS^{(I)}/df$, i.e. $S^{(I)}(f) = \int_{-1/2}^{f} \mathcal{S}(\lambda)\, d\lambda$.

Key results & formulas

Theorem — Aliasing and the Nyquist frequency

If a continuous-time process with SDF $\mathcal{S}_c$ is sampled at $\Delta = 1$, the discrete SDF folds all higher-frequency content onto $[-\frac{1}{2}, \frac{1}{2}]$:

$$\mathcal{S}(f) = \sum_{k\in\mathbb{Z}} \mathcal{S}_c(f + k).$$

The frequency $\frac{1}{2}$ (i.e. $\frac{1}{2\Delta}$) is the **Nyquist frequency**. If $\mathcal{S}_c$ is essentially zero for $|f| \geq \frac{1}{2}$, the discrete and continuous spectra agree well; otherwise the high frequencies corrupt the low ones and $\mathcal{S}$ tells us nothing about $\mathcal{S}_c$.

Aliasing in one image

A high-frequency sinusoid sampled too slowly looks identical to a low-frequency one — the wagon-wheel effect. Frequencies f and $f + k$ become indistinguishable from the samples; they are “aliases”. Cure: sample finely (small Δ, large Nyquist) or low-pass filter before sampling.

Theorem — Herglotz

A complex sequence $\{g_t\}_{t\in\mathbb{Z}}$ is **non-negative definite** *iff* there is a right-continuous, bounded, non-decreasing $G^{(I)}$ with $G^{(I)}(-\frac{1}{2}) = 0$ such that

$$g_t = \int_{-1/2}^{1/2} e^{2\pi itf}\, dG^{(I)}(f), \qquad t \in \mathbb{Z}.$$

Idea: build $G^{(I)}$ as the weak limit of the non-negative Fejér averages of $\sum g_t e^{-2\pi itf}$.

Since every ACVS is non-negative definite, applying Herglotz to $\{\gamma_\tau\}$ shows that **every stationary process has an integrated spectrum** $S^{(I)}$ — even harmonic processes that have no ordinary SDF. The SDF is just the special, absolutely continuous case $dS^{(I)} = \mathcal{S}\, df$.

Theorem — Lebesgue decomposition of $S^{(I)}$

Every integrated spectrum splits uniquely as $S^{(I)} = S_1^{(I)} + S_2^{(I)} + S_3^{(I)}$, each piece non-negative, non-decreasing, $S_j^{(I)}(-\frac{1}{2}) = 0$, where

1. $S_1^{(I)}$ is **absolutely continuous**: $S_1^{(I)}(f) = \int_{-1/2}^{f} \mathcal{S}(\lambda)\, d\lambda$ (the ordinary SDF);
2. $S_2^{(I)}$ is a **step function** with jumps $\{p_\ell\}$ at frequencies $\{f_\ell\}$ of pure sinusoids (a *line spectrum*);
3. $S_3^{(I)}$ is **continuous singular** (pathological, of no practical use).

Reading the three pieces

Absolutely continuous ($S_1^{(I)}$ only): the ACVS damps to zero, $\gamma_\tau \to 0$ — e.g. any ARMA process. **Discrete / line** ($S_2^{(I)}$ only): the ACVS does *not* damp (a sinusoid keeps oscillating) — e.g. the harmonic process. **Mixed**: a sum of the two, e.g. a periodic signal buried in coloured (ARMA) noise.

Theorem — Spectral representation

Let $\{X_t\}$ be real-valued, stationary, mean μ. There is an **orthogonal-increment process** $\{Z(f)\}$ on $[-\frac{1}{2}, \frac{1}{2}]$ with

$$X_t = \mu + \int_{-1/2}^{1/2} e^{2\pi i f t}\, dZ(f) \qquad \text{(equality in mean square)},$$

where, for $f \neq f'$,

$$\mathbb{E}[dZ(f)] = 0, \quad \operatorname{Var}\big(dZ(f)\big) = dS^{(I)}(f), \quad \operatorname{Cov}\big(dZ(f), dZ(f')\big) = 0.$$

The deep structural statement

Every stationary process is a superposition of uncorrelated random sinusoids $e^{2\pi i f t}\, dZ(f)$, carrying "amount of variance" $dS^{(I)}(f)$ at each frequency. This is the precise meaning of the slogan "$\mathcal{S}(f)\, df$ is the power in $[f, f+df]$". (Recall: for complex variables $\operatorname{Cov}(Z_1, Z_2) = \mathbb{E}[(Z_1 - \mathbb{E}Z_1)\overline{(Z_2 - \mathbb{E}Z_2)}]$, with the conjugate on the second factor.)

A tiny worked example

SDF of white noise

Let $\{\varepsilon_t\}$ be white noise with variance σ^2, so $\gamma_\tau = \sigma^2$ at $\tau = 0$ and 0 otherwise. Only the $\tau = 0$ term survives:

$$\mathcal{S}(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau} = \sigma^2 \quad \text{for all } f.$$

The spectrum is **flat** — equal power at every frequency, which is exactly why it is called "white". (Check: $\int_{-1/2}^{1/2} \sigma^2\, df = \sigma^2 = \gamma_0$.)

Key takeaways

Key takeaways — the week's reflexes

- **SDF = Fourier transform of the ACVS** (when $\{\gamma_\tau\} \in \ell^1$). Be able to compute it both ways: $\mathcal{S} = \sum \gamma_\tau e^{-2\pi i f \tau}$ and back $\gamma_\tau = \int \mathcal{S}\, e^{2\pi i f \tau} df$, with $\gamma_0 = \int \mathcal{S} = \operatorname{Var}(X_t)$.
- Use the even/real form $\mathcal{S}(f) = \gamma_0 + 2\sum_{\tau \geq 1} \gamma_\tau \cos(2\pi f \tau)$. Sanity-check any SDF: real, even, ≥ 0, 1-periodic, integrates to γ_0.
- A candidate $\{\gamma_\tau\}$ is a valid ACVS *iff* its transform $\mathcal{S}(f) \geq 0$ for all f; one negative value (often at $f = 0$ or $f = \frac{1}{2}$) disqualifies it.
- Harmonic process $\Rightarrow$ no ordinary SDF, but **Herglotz guarantees an integrated spectrum $S^{(I)}$ always exists**. The SDF is its derivative when it exists.
- Classify a spectrum via Lebesgue: continuous ($\gamma_\tau \to 0$, e.g. ARMA), line/discrete (γ_τ persists, e.g. sinusoid), or mixed.

- Remember Nyquist $= \frac{1}{2}$ and the aliasing fold $\mathcal{S}(f) = \sum_k \mathcal{S}_c(f+k)$: sample fast enough or the high frequencies masquerade as low ones.